Experiment 9.2

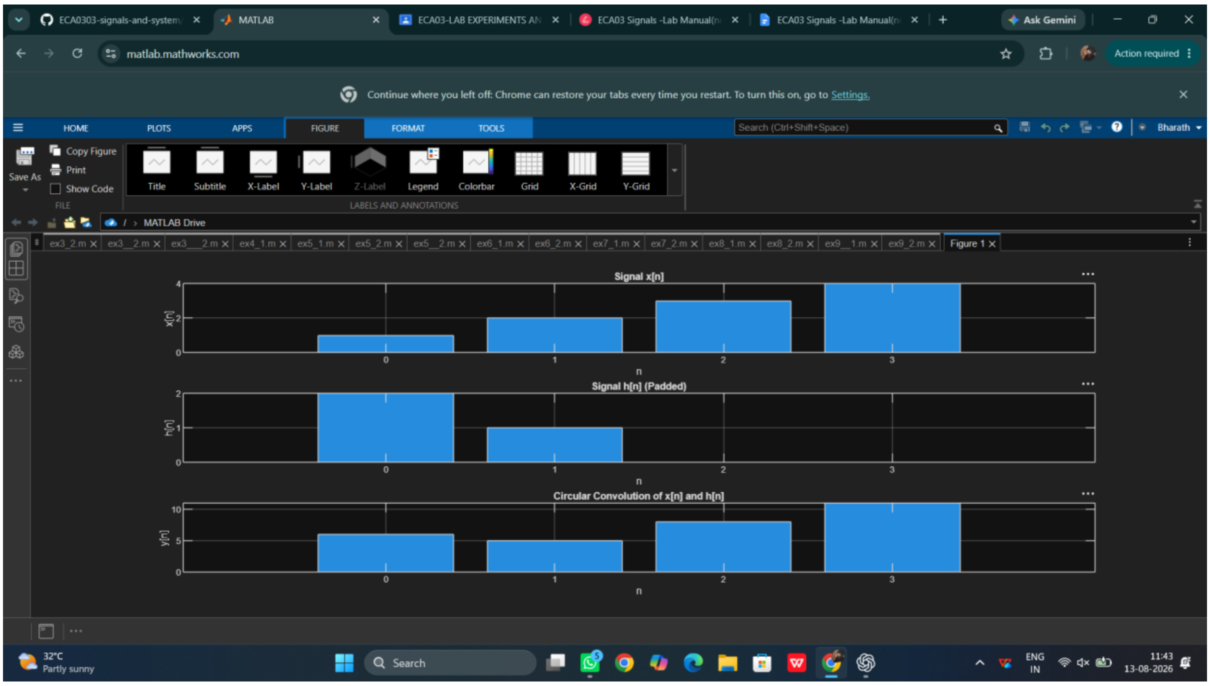
Circular Convolution

Program:

```
// Test Case: Circular Convolution of Discrete-Time Signals
// Define the discrete time signals
x_n = [1, 2, 3, 4];
// Signal x[n]
h_n = [2, 1];
// Signal h[n]
// Zero-pad h[n] to match the length of x[n]
N = length(x_n);
// Length of the signal x[n]
h_n_padded = [h_n, zeros(1, N - length(h_n))]; // Pad h[n] with zeros
// Compute the circular convolution
y_n = cconv(x_n, h_n_padded, N); // Circular convolution using cconv function
// Define the time index for the output signal
n_y = 0:N-1;
// Time index for y[n]
// Plot the input signals and their circular convolution
clf; // Clear the current figure
// Plot x[n]
subplot(3,1,1);
bar(0:length(x_n)-1, x_n, 'b');
xlabel('n');
ylabel('x[n]');
title('Signal x[n]');
```

```
// Plot h[n] (padded)
subplot(3,1,2);
bar(0:N-1, h_n_padded, 'r');
xlabel('n');
ylabel('h[n]');
title('Signal h[n] (Padded)');
// Plot y[n] (circular convolution result)
subplot(3,1,3);
bar(n_y, y_n, 'g');
xlabel('n');
ylabel('y[n]');
title('Circular Convolution of x[n] and h[n]');
```

Output Waveform:



Result:

This test case will help you understand and visualize the circular convolution of discrete-time signals using Scilab.